

BOND VALUATION

A CASE BOOK

STUDENT EDITION • PROBLEMS ONLY

*25 Applied Problems in Bond Pricing, Yield, Duration, Convexity and Immunization
Set in the Indian Fixed-Income Market*

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A Problem Set for MBA / PGDM Programmes in
Financial Management • Fixed Income Securities • Investment Analysis

For pre-class preparation. Contains case situations and required questions only — no solutions, formulas, or worked answers.

Preface and Note on Use

This is the student edition of the Bond Valuation Case Book, containing twenty-five bond-valuation problems organised into eight thematic parts that follow the natural teaching sequence of a Fixed Income Securities or Investment Analysis course: bond value theorems, term structure and forward rates, bond return decomposition, yield to call, duration, interest-rate risk, immunization, and convexity.

Each case is built around a realistic Indian financial-market setting — a bank treasury, an NBFC, a life insurer, a debt mutual fund, a housing finance company, a primary dealer, or a corporate gratuity trust. Every case gives you the institutional context, the learning objective, the case situation with all the data you need, and the required questions to answer.

This edition deliberately contains no solutions, formulas, worked calculations, final answers, interpretations, or teaching notes. It is meant to be read and attempted independently — with a calculator, present-value tables, and the concepts covered in class — before the case is discussed in session.

All figures are stated in Indian Rupees (₹), following the standard Indian numbering convention (lakhs and crores) where the amounts involved warrant it.

Attempt each case in the order presented: the eight parts build on one another, moving from bond pricing fundamentals through to the advanced interest-rate-risk toolkit of duration, immunization and convexity.

Structure of This Case Book

Part	Topic	Cases
Part I	Bond Value Theorems	<i>Cases 1–5</i>
Part II	Term Structure and Forward Rates	<i>Cases 6–8</i>
Part III	Bond Return Decomposition	<i>Cases 9–15</i>
Part IV	Yield to Call	<i>Case 16</i>
Part V	Duration	<i>Cases 17–19</i>
Part VI	Interest-Rate Risk	<i>Cases 20–22</i>
Part VII	Immunization	<i>Case 23</i>
Part VIII	Convexity	<i>Cases 24–25</i>

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Part I — Bond Value Theorems

Five cases establishing the foundational relationships between bond price, coupon rate, yield to maturity, maturity, and the direction and size of yield changes.

Case 1: Discount Narrows as Maturity Approaches

Apex Infrastructure Finance Ltd — Corporate Treasury Desk

Learning Objective

To value a bond trading at a discount to face value and to demonstrate that, when the required yield exceeds the coupon rate, the discount progressively narrows as the bond moves toward maturity.

Case Situation

The corporate treasury desk of Apex Infrastructure Finance Ltd, a Mumbai-based NBFC engaged in infrastructure lending, holds a six-year corporate bond of face value ₹5,000 carrying an annual coupon rate of 9%. Interest rates in the corporate bond market have since hardened, and comparable AA-rated paper is now trading to yield 11%.

The desk head has asked the treasury analyst to mark the bond to market at the current required yield of 11%, and to trace how the bond's price would move each year as it runs down to redemption, assuming the market's required yield of 11% stays unchanged. This exercise feeds into the desk's quarterly mark-to-market note for the investment committee.

Required

1. Calculate the bond's price today (with six years remaining to maturity) at a required yield of 11%.
2. Trace how the bond price changes as the remaining maturity falls from 6 years to 0 years, with YTM held constant at 11%.
3. Explain, in terms of the coupon rate and YTM, why the bond trades below face value and why the gap closes over time.

Case 2: Premium Narrows as Maturity Approaches

Apex Infrastructure Finance Ltd — Corporate Treasury Desk

Learning Objective

To value a bond trading at a premium to face value and to demonstrate that, when the coupon rate exceeds the required yield, the premium progressively narrows as the bond moves toward maturity.

Case Situation

Six months later, the interest-rate environment has reversed. The same 9%-coupon, six-year Apex Infrastructure Finance bond is now being evaluated in a falling-rate scenario: comparable corporate paper is yielding only 7%, well below the bond's coupon.

The treasury analyst has been asked to repeat the earlier exercise — value the bond at the new required yield of 7%, and trace the price path as the bond runs down to maturity — so that the investment committee can see how the bond's premium behaves as rates fall and time passes.

Required

1. Calculate the bond's price today (with six years remaining to maturity) at a required yield of 7%.
2. Trace how the bond price changes as remaining maturity falls from 6 years to 0 years, with YTM held constant at 7%.
3. Explain why the bond now trades above face value and why the premium erodes over time.

Case 3: Maturity and Price Sensitivity to Yield Changes

Konkan Bank Ltd — Treasury and Investments Desk

Learning Objective

To demonstrate that, for a given change in YTM, longer-maturity bonds experience a greater percentage change in price than shorter-maturity bonds — the foundation of interest-rate (duration) risk.

Case Situation

The investments desk at Konkan Bank Ltd holds two AAA-rated corporate bonds in its held-for-trading book, both carrying an 8% coupon and currently priced at par (YTM = 8%): Bond A, maturing in 4 years, and Bond B, maturing in 9 years.

Ahead of the RBI's next monetary policy review, the bank's ALM committee wants to understand how much more the longer bond would fall in value than the shorter one if market yields moved up by just 1 percentage point, to gauge which position carries greater interest-rate risk.

Required

1. Calculate the price of Bond A and Bond B at the current YTM of 8% and at a revised YTM of 9%.
2. Calculate the percentage change in price for each bond.
3. Explain why the longer-maturity bond is more sensitive to the yield change.

Case 4: The Asymmetry of Bond Price Changes

Konkan Bank Ltd — Treasury and Investments Desk

Learning Objective

To demonstrate that, for a given bond, an equal-sized fall in YTM produces a larger percentage price gain than the percentage price loss caused by an equal-sized rise in YTM — the practical face of bond price convexity.

Case Situation

As part of the same ALM stress-testing exercise, Konkan Bank's risk team wants to test a single seven-year, 8%-coupon bond (currently priced at par, since YTM also stands at 8%) under two symmetric yield shocks: a 2-percentage-point rise to 10%, and a 2-percentage-point fall to 6%.

The risk team wants to know whether the bond would gain and lose the same percentage of value under these opposite but equal-sized shocks, since this has a direct bearing on how the bank models tail risk in its interest-rate scenarios.

Required

1. Calculate the bond price and percentage price decrease when YTM rises from 8% to 10%.
2. Calculate the bond price and percentage price increase when YTM falls from 8% to 6%.
3. Compare the magnitude of the two percentage changes and explain the difference.

Case 5: Coupon Rate and Price Sensitivity

Himalaya Debt Advantage Fund — Fixed Income Desk

Learning Objective

To demonstrate that, for a given maturity and a given change in YTM, a higher-coupon bond experiences a smaller percentage price change than a lower-coupon bond.

Case Situation

A fund manager at Himalaya Debt Advantage Fund, an open-ended Indian debt mutual fund, is comparing two five-year corporate bonds of identical face value and maturity for the fund's accrual portfolio: Bond A carries an 8% coupon, and Bond B carries an 11% coupon.

Both bonds currently yield to their respective coupon rates. The fund manager wants to know which bond would lose more value in percentage terms if the market's required yield rose uniformly to 10% for both — information that matters for choosing between the two for a portfolio that may need to be rebalanced before maturity.

Required

1. Calculate the price of Bond A and Bond B at YTM = 8% and at YTM = 10%.
2. Calculate the percentage change in price for each bond.
3. Explain why the higher-coupon bond is less sensitive to the change in yield.

Part II — Term Structure and Forward Rates

Three cases on extracting forward rates from the spot-rate curve, and on the classic yield-curve-riding strategy for money-market instruments.

Case 6: Extracting Forward Rates from the G-Sec Spot Curve

Primary Dealer Desk, Government Securities Trading Desk

Learning Objective

To derive the market's implied one-year forward rates from a set of observed spot yields on Central Government Securities, using the no-arbitrage relationship between spot and forward rates.

Case Situation

The government securities trading desk of a primary dealer is building its term-structure model for the coming quarter. The desk observes the following spot yields on Central Government Securities: the 1-year security yields 7.20%, the 2-year security yields 7.80%, and the 3-year security yields 8.60%.

Before quoting forward-starting swaps and advising institutional clients on curve positioning, the desk's junior dealer has been asked to back out the one-year forward rates implied by this spot curve — the rates the market is effectively pricing for future one-year borrowing and lending.

Required

1. Calculate the one-year forward rate beginning one year from now, f_2 .
2. Calculate the one-year forward rate beginning two years from now, f_3 .

Case 7: The Implied Forward Rate for Year 5

Setu Housing Finance Corp Ltd — Corporate Treasury

Learning Objective

To derive a single implied forward rate from just two spot rates, using the shortcut ratio method — a common requirement when a firm is deciding on the timing of future borrowing.

Case Situation

The corporate treasury of Setu Housing Finance Corp Ltd is evaluating the cost of a bullet borrowing it plans to raise four years from now, to be repaid one year later (i.e., a one-year loan starting in year 5). The treasury team observes that five-year bonds in the market currently yield 7.6%, while four-year bonds yield 6.8%.

Before approaching lenders, the treasurer wants to estimate the one-year borrowing rate the market is implicitly pricing for that future window, so the company can judge whether locking in a forward-rate agreement today would be cheaper than waiting and borrowing at the prevailing spot rate in four years.

Required

1. Calculate the implied one-year forward rate, f_5 , for a one-year investment/borrowing beginning four years from now.

Case 8: Riding an Upward-Sloping Yield Curve with Treasury Bills

Himalaya Debt Advantage Fund — Money Market Desk

Learning Objective

To evaluate the 'riding the yield curve' strategy — buying a longer-dated money-market instrument and selling it before maturity to capture price appreciation from rolling down an upward-sloping curve.

Case Situation

The money-market desk of Himalaya Debt Advantage Fund is deciding how to deploy short-term surplus cash. Two T-bills are quoted: a 91-day T-bill priced at ₹97.80 and a 182-day T-bill priced at ₹95.20 (both per ₹100 face value).

Rather than simply buying and holding the 91-day bill to its own maturity, the desk is considering the classic 'riding the yield curve' trade: buy the 182-day bill today, and sell it after 91 days — at which point it will have exactly 91 days left to run, and should, if the yield curve is unchanged, be priced like a fresh 91-day bill, i.e. at ₹97.80.

Required

1. Calculate the investor's annualised holding-period yield from riding the curve (buying the 182-day bill and selling it after 91 days at an assumed price of ₹97.80).
2. Explain the strategy and the condition under which it works.

Part III — Bond Return Decomposition

Seven linked cases building up an investor's total bond return — coupon income, interest-on-interest, and capital gain — and converting it into a realized yield across different holding periods and reinvestment-rate scenarios.

Case 9: Interest-on-Interest and the Reinvestment Rate

Suraksha Life Insurance Co. Ltd — Fixed Income Portfolio

Learning Objective

To compute the interest-on-interest (Iol) component of a bond's total return, and to show how it shrinks when the reinvestment rate available on coupon receipts falls below the bond's own yield to maturity.

Case Situation

Suraksha Life Insurance Co. Ltd, a mid-sized Indian life insurer, holds a 12% annual-coupon corporate bond of face value ₹500 in its policyholders' fund. The bond is redeemable after six years at a 4% premium (i.e., at ₹520), and was purchased at ₹470.23, giving it a yield to maturity of approximately 14%.

The insurer's investment team reinvests every coupon it receives in similar short-tenor instruments. Initially, coupons can be reinvested at 14%, matching the bond's own YTM. The team now wants to test a scenario in which the interest-rate environment eases and the reinvestment rate available on incoming coupons falls to 11%, to understand how much of the bond's total return this reinvestment risk could take away by the time the bond matures in six years.

Required

1. Calculate the interest-on-interest (Iol) earned over the bond's six-year life if coupons are reinvested at 14%.
2. Calculate the Iol if coupons are instead reinvested at 11%.
3. Compare the two results and comment on the effect of a falling reinvestment rate.

Case 10: Interest-on-Interest Across Different Holding Periods

Suraksha Life Insurance Co. Ltd — Fixed Income Portfolio

Learning Objective

To show how the interest-on-interest component builds up as the holding period lengthens, and how this build-up differs at two different reinvestment rates.

Case Situation

Continuing with the same bond, Suraksha Life Insurance's investment team wants a fuller picture: rather than looking only at the full six-year Iol figure, they want to see how the Iol accumulates year by year, for both the 14% and 11% reinvestment scenarios, so the insurer can assess exposure to reinvestment risk at each potential exit horizon.

Required

1. Determine the interest-on-interest component for holding periods of 1, 2, 3, 4, 5 and 6 years, at reinvestment rates of 14% and 11%.
2. Comment on how Iol behaves as the holding period lengthens.

Case 11: Capital Gain from Selling a Bond Before Maturity*Suraksha Life Insurance Co. Ltd — Fixed Income Portfolio***Learning Objective**

To compute the market price and resulting capital gain if the bond is sold before maturity, and to show how this capital gain builds up as the holding period lengthens, assuming the required return stays at 14%.

Case Situation

Suraksha Life Insurance Co. Ltd, a mid-sized Indian life insurer, holds a 12% annual-coupon corporate bond of face value ₹500 in its policyholders' fund. The bond is redeemable after six years at a 4% premium (i.e., at ₹520), and was purchased at ₹470.23, giving it a yield to maturity of approximately 14%.

If held to maturity, the bond's redemption value of ₹520 is already known with certainty. But insurers frequently rebalance portfolios before maturity, and the investment team wants to know what the bond would fetch — and what capital gain it would represent over the ₹470.23 purchase price — if sold at various points before the six-year maturity, assuming the required return in the market stays at 14%.

Required

1. Calculate the market price of the bond after one year, assuming the required return remains 14%.
2. Calculate the resulting capital gain over the ₹470.23 purchase price.
3. Extend the calculation to holding periods of 2, 3, 4, 5 and 6 years, and comment on the pattern.

Case 12: Capital Gain When the Required Return Falls*Suraksha Life Insurance Co. Ltd — Fixed Income Portfolio***Learning Objective**

To show how a fall in the market's required return magnifies the capital gain available from selling a bond before maturity, illustrating the inverse relationship between yields and bond prices.

Case Situation

Using the same bond, the investment team now asks: what if interest rates fall and the required return applicable to this bond drops from 14% to 11%? The team wants the market price and capital gain recalculated for holding periods of one to six years under this lower required return, to compare against the 14% scenario already computed in Case 11.

Required

1. Calculate the market price and capital gain for holding periods of 1 to 6 years, assuming the required return is 11%.
2. Compare the results with the 14% scenario and explain the difference.

Case 13: Total Return: Coupon, Interest-on-Interest and Capital Gain Combined

Suraksha Life Insurance Co. Ltd — Fixed Income Portfolio

Learning Objective

To assemble the three components of bond return — coupon income, interest-on-interest, and capital gain — into a single total-return figure, and to show how the components trade off against each other under different reinvestment/required-return scenarios.

Case Situation

Having computed interest-on-interest (Cases 9-10) and capital gain (Cases 11-12) separately, the investment team at Suraksha Life Insurance now wants to see the complete picture: the total rupee return generated by the bond over each holding period, under both the 14% and 11% rate scenarios, so that the two scenarios can be compared on a like-for-like basis.

Required

1. Assemble the total return (coupon income + interest-on-interest + capital gain) for holding periods of 1 to 6 years, at a reinvestment/required rate of 14%.
2. Repeat the exercise at a reinvestment/required rate of 11%.
3. Explain the trade-off visible between the two scenarios.

Case 14: Realized Yield for a One-Year Holding Period

Suraksha Life Insurance Co. Ltd — Fixed Income Portfolio

Learning Objective

To compute the realized (holding-period) yield on the bond — the actual annualised rate of return earned by the investor — and to show why it can differ sharply from both the bond's YTM and the prevailing reinvestment rate.

Case Situation

Using the total-return figures already computed in Case 13, the investment team at Suraksha Life Insurance wants to convert the one-year total return under each rate scenario into a realized yield — a single percentage figure comparable to the bond's original YTM of 14% — to judge how the insurer actually fared over that year.

Required

1. Calculate the realized yield for a one-year holding period, assuming a 14% reinvestment/required rate.

2. Calculate the realized yield for a one-year holding period, assuming an 11% reinvestment/required rate.
3. Explain why the second figure is so much higher than the reinvestment rate that produced it.

Case 15: Realized Yield Across a Full Range of Holding Periods

Suraksha Life Insurance Co. Ltd — Fixed Income Portfolio

Learning Objective

To trace how realized yield evolves as the holding period lengthens from 1 to 6 years, and to demonstrate that realized yield converges to YTM only when the reinvestment rate equals YTM, or as the holding period approaches maturity.

Case Situation

To complete the analysis, Suraksha Life Insurance's investment team wants the realized-yield calculation extended across all six possible holding periods (1 through 6 years), for both the 14% and 11% rate scenarios, drawing on the total-return figures already built up in Case 13.

Required

1. Determine the realized yield for holding periods of 1 to 6 years under a 14% reinvestment/required rate.
2. Determine the realized yield for holding periods of 1 to 6 years under an 11% reinvestment/required rate.
3. Explain the pattern observed in the 11% scenario as the holding period lengthens.

Part IV — Yield to Call

One case on valuing a callable bond to its call date rather than its final maturity.

Case 16: Yield to Call on a Callable Corporate Bond

Himalaya Debt Advantage Fund — Fixed Income Desk

Learning Objective

To compute the Yield to Call (YTC) of a callable bond — the return earned if the bond is redeemed at the issuer's option on the call date, rather than held to final maturity.

Case Situation

Himalaya Debt Advantage Fund holds a bond issued by Bharat Renewable Energy Ltd, a wind and solar power generation company. The bond has a face value of ₹1,000 and carries an 11% coupon paid semiannually, with a final maturity of eight years. However, the issuer holds a call option: it may redeem the bond at the end of four years, at a call premium of 8% over face value.

The bond currently trades in the market at ₹1,040. Given the coupon is well above prevailing market rates, the fund's analyst believes the issuer is highly likely to call the bond at the four-year mark rather than let it run to full maturity, and wants to compute the yield the fund would actually earn under that call scenario.

Required

1. Calculate the bond's yield to call (YTC), assuming the bond is called at the end of year 4 at the stated call premium.

Part V — Duration

Three cases computing Macaulay duration — by direct cash-flow weighting, by the coupon-rate/YTM shortcut formula, and its conversion into modified duration.

Case 17: Macaulay Duration from Discounted Cash Flows

Suraksha Life Insurance Co. Ltd — Fixed Income Portfolio

Learning Objective

To compute a bond's Macaulay duration directly from its schedule of discounted cash flows, and to show how the reinvestment/discount rate used affects the resulting duration figure.

Case Situation

Suraksha Life Insurance's risk team wants to measure the interest-rate exposure of the same 12% coupon, ₹500 face-value bond used throughout Part III (Cases 9-15), by computing its Macaulay duration — the present-value-weighted average time to receipt of its cash flows — at both the 14% and 11% discount rates already used in the earlier analysis.

Required

1. Using the bond's cash flows (₹60 in years 1-5 and ₹580 in year 6, comprising the final coupon plus redemption value), compute Macaulay duration at a 14% discount rate.
2. Repeat the computation at an 11% discount rate, and comment on the difference.

Case 18: Duration Using the Coupon-Rate/YTM Shortcut Formula

Konkan Bank Ltd — Treasury and Investments Desk

Learning Objective

To compute Macaulay duration using the closed-form shortcut formula linking a bond's current yield to its YTM, and to note the special case where the bond is priced at par.

Case Situation

Konkan Bank's treasury desk holds a corporate bond with a face value of ₹2,000, a 7% annual coupon, and five years remaining to maturity, priced to yield 9%. Rather than build out the full cash-flow table each time, the desk's quant analyst wants to apply the algebraic shortcut formula for duration, which uses only the bond's current yield and YTM, to speed up daily duration reporting across the desk's larger bond book.

Required

1. Calculate the bond's price and current yield at the stated YTM of 9%.
2. Using the shortcut duration formula, calculate the bond's Macaulay duration.
3. State how the formula simplifies when a bond is priced at par.

Case 19: Modified Duration and Estimated Price Sensitivity

Himalaya Debt Advantage Fund — Risk Management Desk

Learning Objective

To convert Macaulay duration into modified duration, and to use it to estimate the percentage change in a bond's price for a given change in YTM — the single most widely used measure of interest-rate risk in practice.

Case Situation

The risk desk at Himalaya Debt Advantage Fund holds a 12-year, 9%-coupon government security (face value ₹1,00,000) priced at par (YTM = 9%). Ahead of a scheduled RBI policy announcement, the desk wants a quick, standard estimate of how much the bond's price would move if YTM rose by 1 percentage point — the kind of number quoted routinely on a bond desk's daily risk report.

Required

1. Calculate the bond's Macaulay duration, using the par-bond shortcut from Case 18.
2. Calculate the bond's modified duration.
3. Estimate the percentage change in the bond's price for a 1-percentage-point increase in YTM.

Part VI — Interest-Rate Risk

Three cases measuring bond price sensitivity to yield changes — interest rate elasticity, its link to duration, and effective duration computed from directly repriced bonds.

Case 20: Interest Rate Elasticity of a Par Bond

Himalaya Debt Advantage Fund — Risk Management Desk

Learning Objective

To measure a bond's interest rate elasticity — the percentage change in price for a given percentage change in YTM — as a scale-free alternative to quoting price sensitivity in absolute terms.

Case Situation

Continuing its RBI-announcement risk assessment, Himalaya Debt Advantage Fund's risk desk wants to express the same 12-year, 9%-coupon, par-priced government security's interest-rate risk not just as a rupee or percentage price move, but as an elasticity figure — directly comparable, in principle, to the price elasticities used elsewhere in finance and economics.

Required

1. Calculate the bond's interest rate elasticity if YTM rises from 9% to 10%.

Case 21: Linking Duration to Interest Rate Elasticity

Himalaya Debt Advantage Fund — Risk Management Desk

Learning Objective

To derive interest rate elasticity directly from a bond's duration, demonstrating the algebraic link between the two measures of interest-rate risk.

Case Situation

Having computed interest rate elasticity directly from price changes in Case 20, the risk desk now wants to verify the figure using an entirely separate route — computing it from the bond's duration — both as a cross-check and to build the desk's intuition for how duration and elasticity relate.

Required

1. Calculate the duration of the same 12-year, 9%-coupon government security (priced at par).
2. Use the duration-elasticity relationship to compute interest rate elasticity, and compare it with the directly computed figure from Case 20.

Case 22: Effective Duration from an Observed Yield Shock

*Konkan Bank Ltd — Treasury and Investments Desk***Learning Objective**

To compute effective duration directly from observed bond prices under a small up- and down-shock to yield, a method that remains valid even when a bond's cash flows themselves may change with yield.

Case Situation

Konkan Bank's treasury desk holds an eight-year, 7.5%-coupon bond (face value ₹1,000), currently priced at par since its YTM also stands at 7.5%. As part of a routine risk-model validation, the desk's quant analyst reprices the bond under a small parallel yield shock of 25 basis points in each direction — down to 7.25% and up to 7.75% — and wants to derive the bond's effective duration directly from the resulting prices, rather than from a formula based on the bond's stated cash flows.

Required

1. Given that the bond reprices to ₹1,014.78 when the yield falls to 7.25%, and to ₹985.50 when the yield rises to 7.75%, calculate the bond's effective duration.

Part VII — Immunization

One case applying duration matching to immunize a stream of future liabilities using a two-bond zero-coupon portfolio.

Case 23: Immunizing a Five-Year Gratuity Payout

Approved Gratuity Trust of an Indian Manufacturing Company

Learning Objective

To compute the duration of a stream of fixed future liabilities, and to determine how a portfolio of zero-coupon bonds should be allocated so that the assets' duration matches the liabilities' duration — the core technique of duration-based immunization.

Case Situation

The trustees of the approved gratuity trust of an Indian manufacturing company must fund a retiring employee's gratuity, to be paid out in five equal annual instalments of ₹50,000 each, starting one year from now. The trustees want to invest the corpus today in a way that is protected (immunized) against future interest-rate movements, rather than simply matching the final payment date.

Two zero-coupon government bonds are available for the trust to invest in: one maturing in 2 years, and one maturing in 6 years. The applicable required rate of return on the corpus is 10%.

Required

1. Calculate the duration of the liability stream (the five annual ₹50,000 payments).
2. Determine what percentage of the corpus should be invested in the 2-year zero-coupon bond and what percentage in the 6-year zero-coupon bond so that the portfolio's duration matches the liability duration.

Part VIII — Convexity

Two cases computing bond convexity and using the duration-plus-convexity approximation to improve on duration-only price-sensitivity estimates for large yield moves.

Case 24: Computing Bond Convexity for a High-Coupon Bond

Himalaya Debt Advantage Fund — Fixed Income Desk

Learning Objective

To compute a bond's convexity — the second-order measure of price sensitivity that captures the curvature of the price-yield relationship left unexplained by duration alone.

Case Situation

Himalaya Debt Advantage Fund holds a five-year, 12%-coupon corporate bond (face value ₹2,000), currently priced to yield 9%. Having already used modified duration to estimate this bond's price sensitivity, the fund's quant analyst now wants to compute its convexity, so that a more accurate, curvature-adjusted price-sensitivity estimate can be produced for larger yield moves (a technique applied in Case 25).

Required

1. Calculate the convexity of the bond.

Case 25: Duration-Convexity Approximation for a Large Yield Move

Suraksha Life Insurance Co. Ltd — Long-Duration G-Sec Portfolio

Learning Objective

To compare a duration-only price estimate against a duration-plus-convexity estimate for a large change in yield, and to show how much more accurate the convexity-adjusted estimate is relative to the bond's true, fully revalued price.

Case Situation

Suraksha Life Insurance holds a long-dated, 25-year, 7.5%-coupon government security in its policyholders' fund, purchased at par (face value ₹1,00,000) when its YTM also stood at 7.5%. Following a sharp tightening in the interest-rate environment, the bond's YTM has since risen to 9.5% — a large, 2-percentage-point move that the insurer's ALM desk wants to analyse carefully, since large yield swings are exactly where a simple duration-only estimate is known to be least reliable.

The bond's modified duration at its original 7.5% yield is 11.15 years, and its convexity is 195.56. The desk wants to compare what duration alone would have predicted, what duration-plus-convexity predicts, and how both compare with the bond's actual, fully revalued price at the new 9.5% yield (₹81,124.84).

Required

- 1.** Estimate the percentage decline in the bond's price using modified duration alone.
- 2.** Estimate the percentage decline using the duration-plus-convexity approximation.
- 3.** Compare both estimates against the bond's actual percentage price decline.